

	Sales Document Voltage Control Operation	K0821_045435_EN_R04 Revision 04 17.06.2019
---	---	---

Sales Document

Voltage Control Operation

Document number:

K0821_045435_EN_R04

Revision:

04

Author:

T. Dibbern / ESW

Date:

17.06.2019

Responsible department:

ESW

Checked:

K. Kruppa / ESW

Classification:

Nordex internal Purpose

AST:

21537

Released:

U. Joehnke / ESW

Replaces:

K0821_045435_EN_R03

Validity:

K	Main Assy	Assy Gen	P/T
K08			P
D4K		Delta4000	

Document is published in electronic form. Signed original with Central Engineering/ENS.

© 2019 Nordex Energy GmbH, Langenhorner Chaussee 600, D22419 Hamburg

All rights reserved. Observe protection notice ISO 16016

	Sales Document Voltage Control Operation	K0821_045435_EN_R04 Revision 04 17.06.2019
---	---	---

Document is published in electronic form. Signed original with Central Engineering/ENS.

© 2019 Nordex Energy GmbH, Langenhorner Chaussee 600, D22419 Hamburg

All rights reserved. Observe protection notice ISO 16016

	Sales Document Voltage Control Operation	K0821_045435_EN_R04 Revision 04 17.06.2019
---	---	---



Voltage Control Operation

in the Wind Farm

© 2019 Nordex Energy GmbH, Langenhorner Chaussee 600, D22419 Hamburg
All rights reserved. Observe protection notice ISO 16016

	Sales Document Voltage Control Operation	K0821_045435_EN_R04 Revision 04 17.06.2019
---	---	---

Revision index

Revision	Modification (chapter)	AST
00	First issue	7525
01	Correction & Validity extended on Delta Generation	8934
02	Update for BTF CWE 7.0	20300
03	Correction of hysteresis diagram (Chapter 2.4)	20300
04	Validity extended for Delta4000	21537

Referenced documents

No.	Document number	Description
		NORDEX documents
[1]	K0801_004439_EN	Reactive Power Capability K08
[2]	E0004514776	Reactive Power Capability Delta4000
[3]		
		Standards/guidelines
[4]		
[5]		
[6]		
[7]		

Table of Contents

1	Introduction	3
1.1	General.....	3
1.2	Abbreviations, terms, definitions, symbols.....	3
1.3	The principles of voltage control	3
2	Functional description of the $Q=f(U)$ -function.....	5
2.1	Variable slope.....	6
2.2	Reference Voltage.....	8
2.3	Deadband	9
2.4	Hysteresis.....	11
2.5	Step function	12
2.6	Reactive Power Capability	13
2.7	Set point Changes.....	13
3	Measurement and control accuracy	14
3.1	Dynamic control accuracy	15

	Sales Document Voltage Control Operation	K0821_045435_EN_R04 Revision 04 17.06.2019
---	---	---

3.2	Static measurement deviation.....	15
3.3	Interfaces.....	15
4	Protection Notice ISO 16016.....	15

1 Introduction

1.1 General

This document describes the voltage control of wind farms during standard operation. The term voltage control operation is understood to indicate voltage dependant reactive power closed loop control or reactive power open loop control.

The types of operation and operational measurement- and control deviations will also be described.

In order to evaluate the stability of the control and the adjustment of the voltage controller, a load flow calculation will be required.

1.2 Abbreviations, terms, definitions, symbols

CWE: Combined Wind Farm Management and Electrical Site

Q, P: Reactive Power, Active Power

PT1: 1st Order Delay Element

V: Voltage

WTG: Wind turbine generator

WF: Wind Farm

1.3 The principles of voltage control

The voltage control described here is a voltage dependant reactive power control. The CWE measures the voltage at the point of connection. A reactive power set point is generated via a project specific parameterised $Q=f(U)$ -function dependant on the measured voltage and the voltage set point. This reactive power set point is specified for the voltage controller, which uses the reactive power as the controlled variable

The CWE measures the reactive power at the point of connection and adjusts the difference to the set point. The control variable Q_{con} is distributed via the set point distributor and individual set points are transmitted to each WT. The set point distributor can distribute the reactive power set point equally for all WTs or based on the individual WT's available reactive power.

The function is illustrated in the figure below:

	Sales Document Voltage Control Operation	K0821_045435_EN_R04 Revision 04 17.06.2019
---	---	---

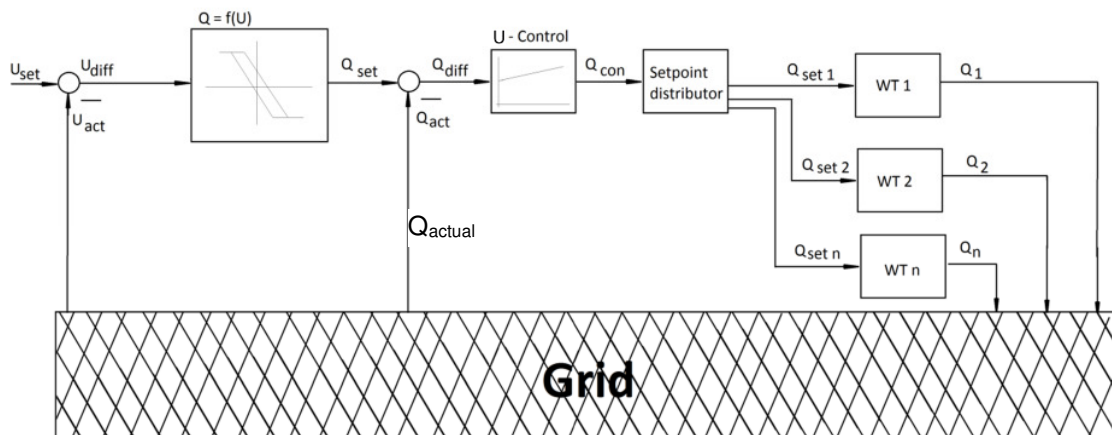


Figure 1: Schema of the voltage control

In this figure, “ U_{set} ” is the current voltage set point and “ U_{act} ” is the measured voltage. “ Q_{set} ” is the calculated Q-set point from the project specific parameterised $Q=f(U)$ -function. Q_{act} is the currently measured reactive power, “ Q_{diff} ” is the control deviation for the U-controller, “ Q_{con} ” is the U-controller output, “ $Q_{set\ 1-n}$ ” are the Q-set points distributed to the WTs and “ Q_{1-n} ” are the reactive powers produced by the WTs.

When the CWE reactive power controller works in open loop operation, the reactive power is not adjusted at the point of connection. However, the $Q=f(U)$ -function can still be used. The set point resulting from the $Q=f(U)$ -function is transferred directly to the set point distributor.

	Sales Document Voltage Control Operation	K0821_045435_EN_R04 Revision 04 17.06.2019
---	---	---

2 Functional description of the Q=f(U)-function

The reactive power set point calculated by the Q=f(U)-function is based on the deviation between the voltage set point U_{set} and the actual voltage U_{act} .

The gradient of the function can be set internally or can be set externally via set point. There are two options to set the gradient.

The gradient can be set via ratio k_p :

$$k_p \left[\frac{Mvar}{kV} \right]$$

The gradient can be set via a slope. The CWE converts the slope into a ratio k_p :

$$\text{Slope}[\%] = \frac{Q[Mvar] * 100\%}{\text{reference voltage}[kV] * k_p \left[\frac{Mvar}{kV} \right]}$$

The internal slope and the internal ratio k_p can be set independently for undervoltage and overvoltage. The external slope and the external ratio k_p can only be set for the whole range.

The value “ Q_{min} ” is the minimum reactive power and “ Q_{max} ” is the maximum reactive power of the Q=f(U)-function.

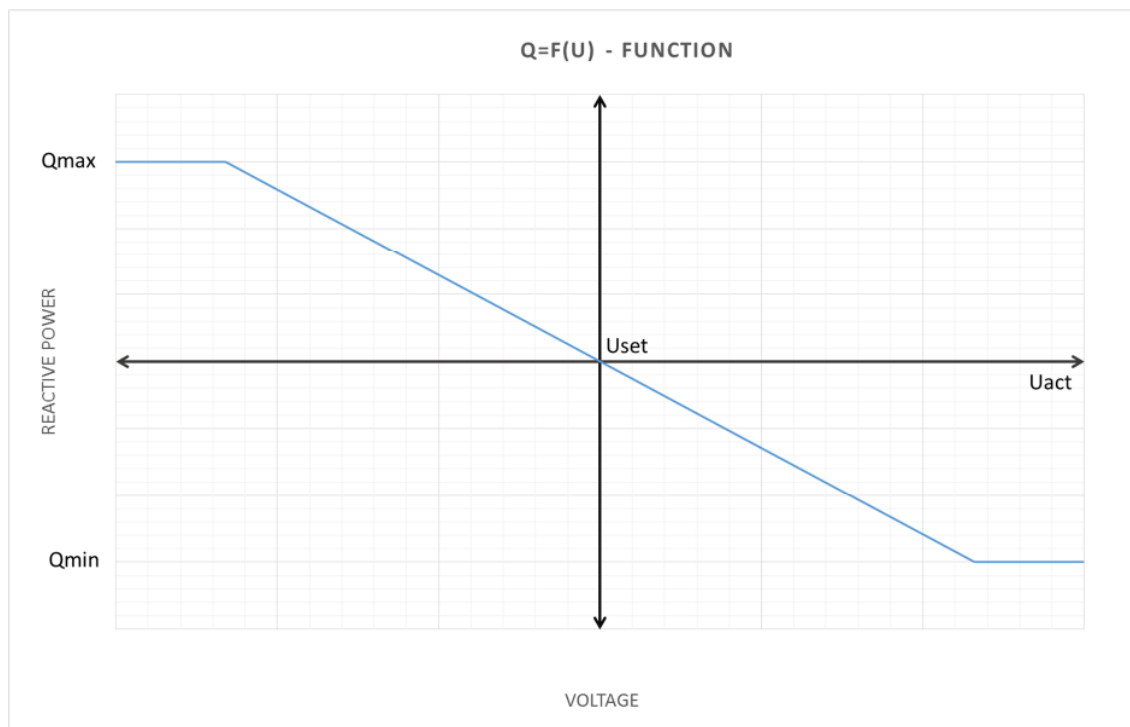


Figure 2: Reactive power set point dependent on the voltage deviation

	Sales Document Voltage Control Operation	K0821_045435_EN_R04 Revision 04 17.06.2019
---	---	---

2.1 Variable slope

If the gradient is defined by a slope, there are two configurable options for the gradient when the available reactive power is lower than the nominal reactive power. The gradient can be constant or dynamic at lower available reactive power:

Constant gradient:

$$k_p \left[\frac{Mvar}{kV} \right] = \frac{Q_{nom}[Mvar] * 100\%}{\text{reference voltage}[kV] * \text{Slope}[\%]}$$

Dynamic gradient:

$$k_p \left[\frac{Mvar}{kV} \right] = \frac{Q_{avail}[Mvar] * 100\%}{\text{reference voltage}[kV] * \text{Slope}[\%]}$$

When the gradient is defined via k_p the gradient will always be constant.

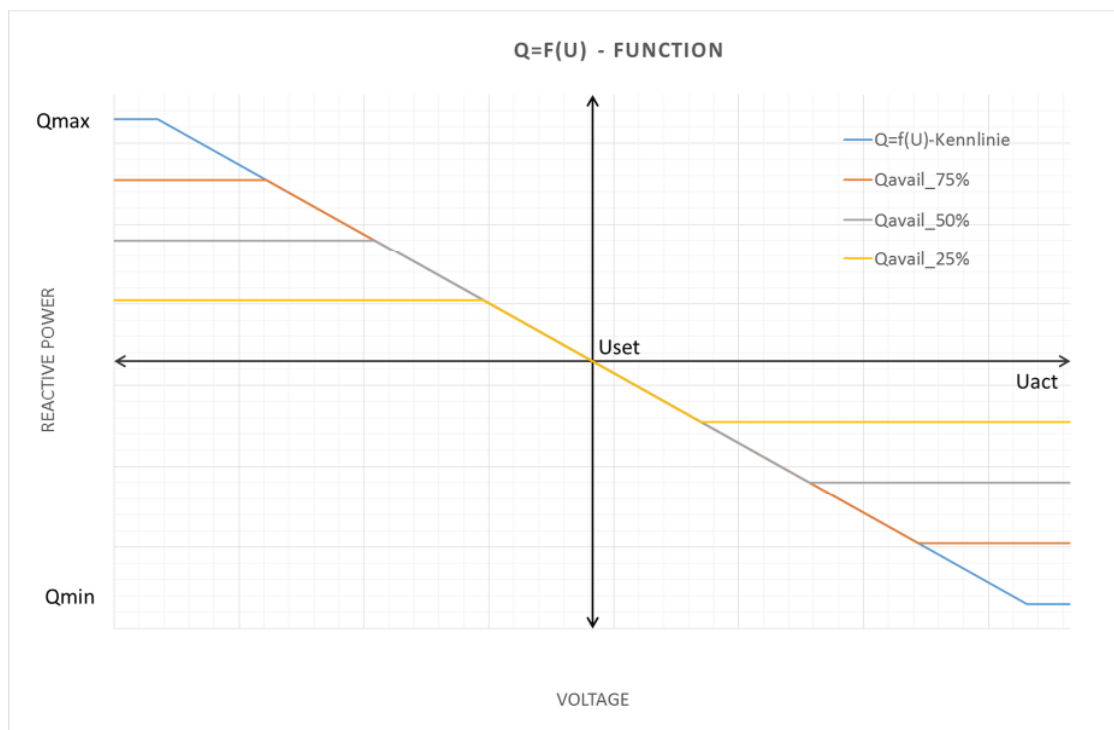


Figure 3: Constant gradient

	Sales Document Voltage Control Operation	K0821_045435_EN_R04 Revision 04 17.06.2019
---	---	--

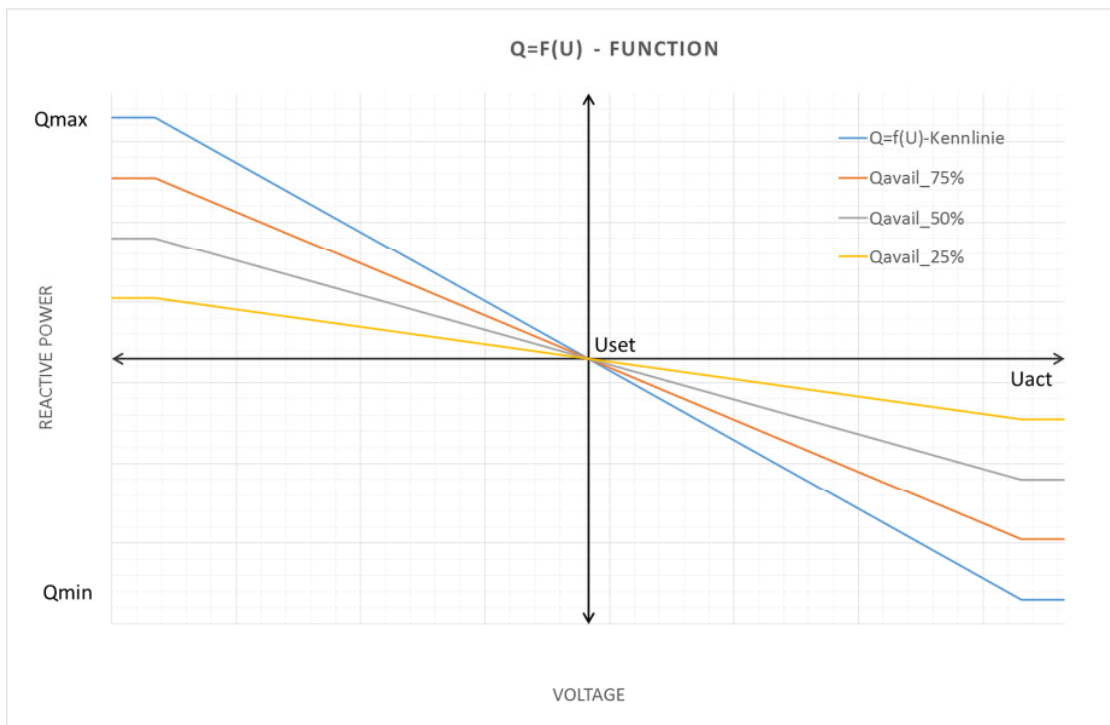


Figure 4: Dynamic gradient

	Sales Document Voltage Control Operation	K0821_045435_EN_R04 Revision 04 17.06.2019
---	---	---

2.2 Reference Voltage

When the gradient is defined via slope the CWE converts the slope into a ration k_p with the formula:

$$k_p \left[\frac{Mvar}{kV} \right] = \frac{Q[Mvar] * 100\%}{\text{reference voltage}[kV] * \text{Slope}[\%]}$$

The reference voltage used for this calculation can be configured. When the nominal voltage is used for the calculation the gradient remains constant for every voltage setpoint. When the voltage setpoint itself is used for the calculation of the gradient, the gradient changes dynamically with different voltage set points.

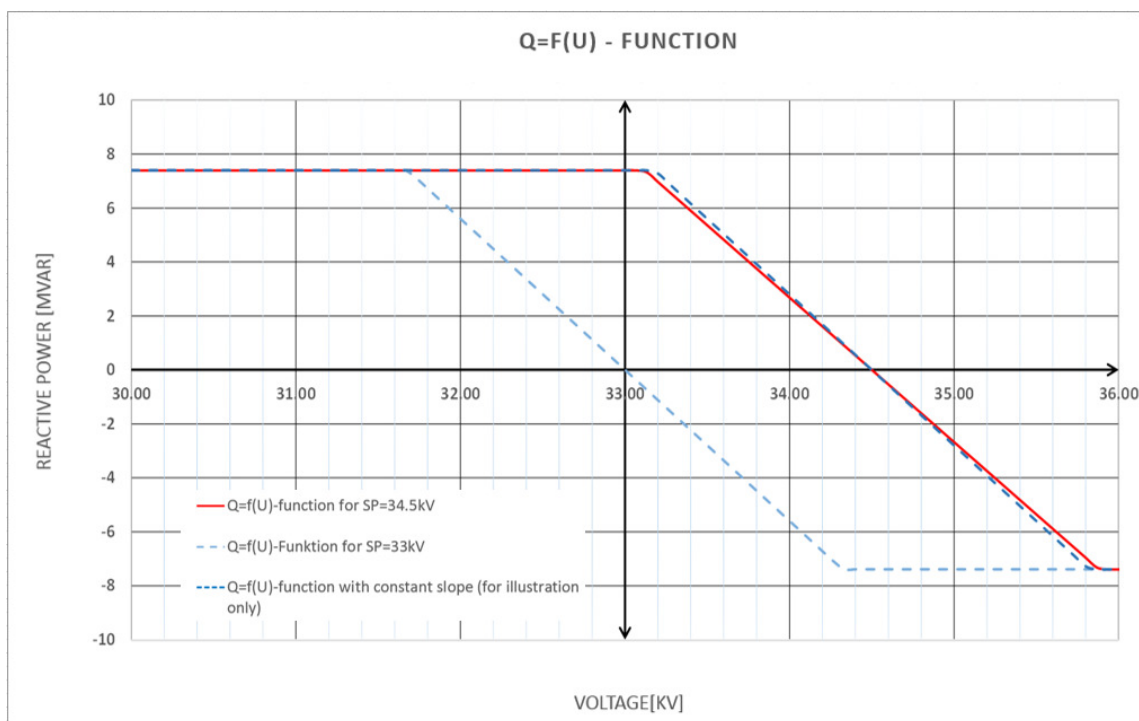


Figure 5: Comparison of constant and dynamic gradient resulting from a different reference voltage.

	Sales Document Voltage Control Operation	K0821_045435_EN_R04 Revision 04 17.06.2019
---	---	---

2.3 Deadband

When the measured voltage is in the configured deadband the reactive power set point is 0kvar. Changes in the measured do not result in changes of the reactive power set point. With an external deadband the deadband will be equally divided for undervoltage and overvoltage (see figure 3). With internal parameters the deadband can be set asymmetrically for undervoltage and overvoltage (see figure 4).

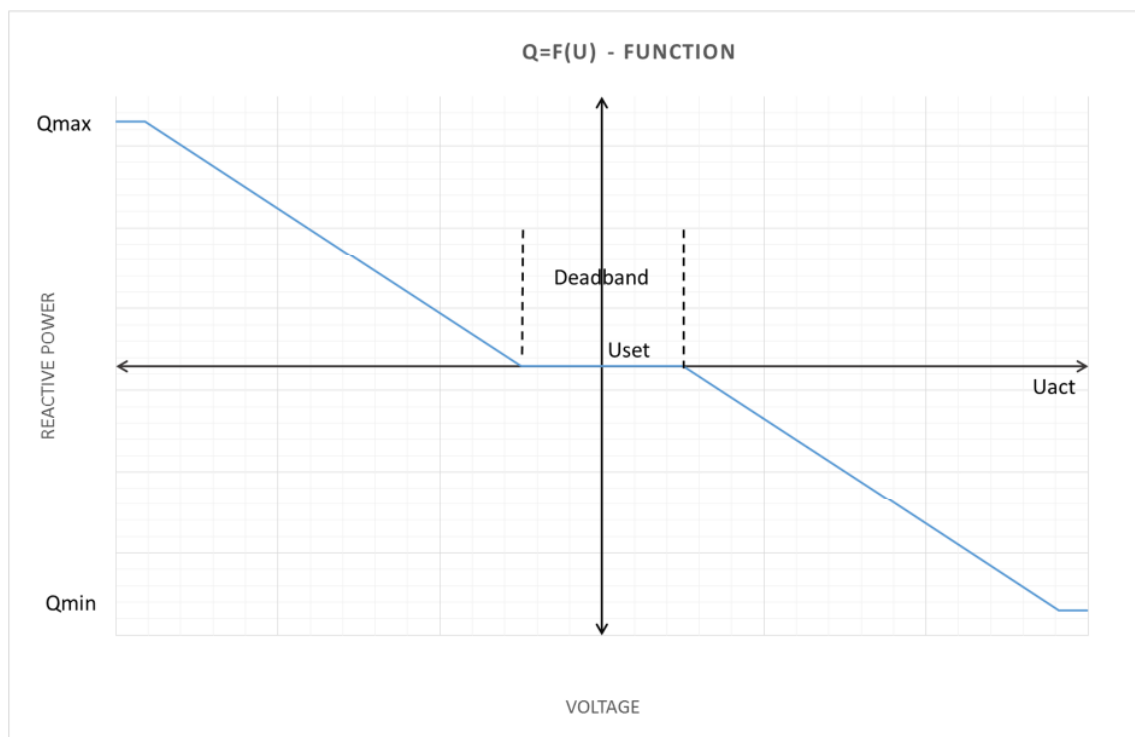


Figure 6: Symmetric deadband

	Sales Document Voltage Control Operation	K0821_045435_EN_R04 Revision 04 17.06.2019
---	---	--

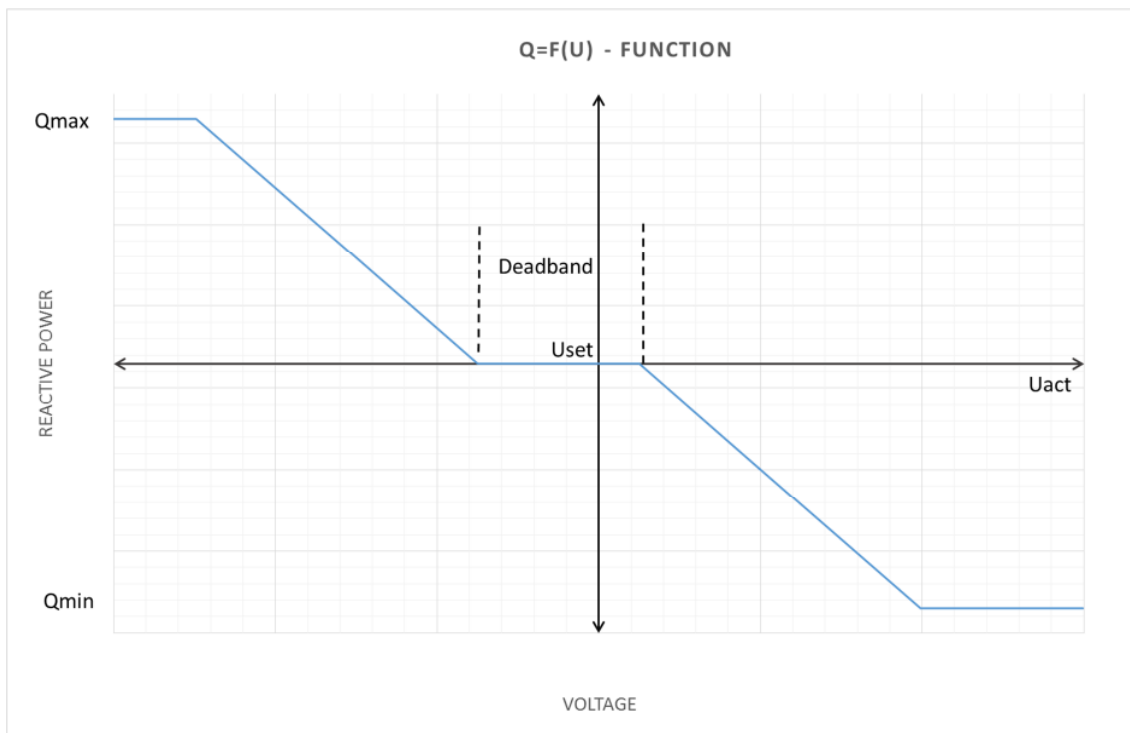


Figure 7: asymmetric deadband

	Sales Document Voltage Control Operation	K0821_045435_EN_R04 Revision 04 17.06.2019
---	---	---

2.4 Hysteresis

A hysteresis can be configured to decrease the dynamic of the $Q=f(U)$ -function. In the range of the hysteresis the reactive power set point of the $Q=f(U)$ -function is kept constant.

The hysteresis must be smaller than half the deadband as ambiguous statuses may occur.

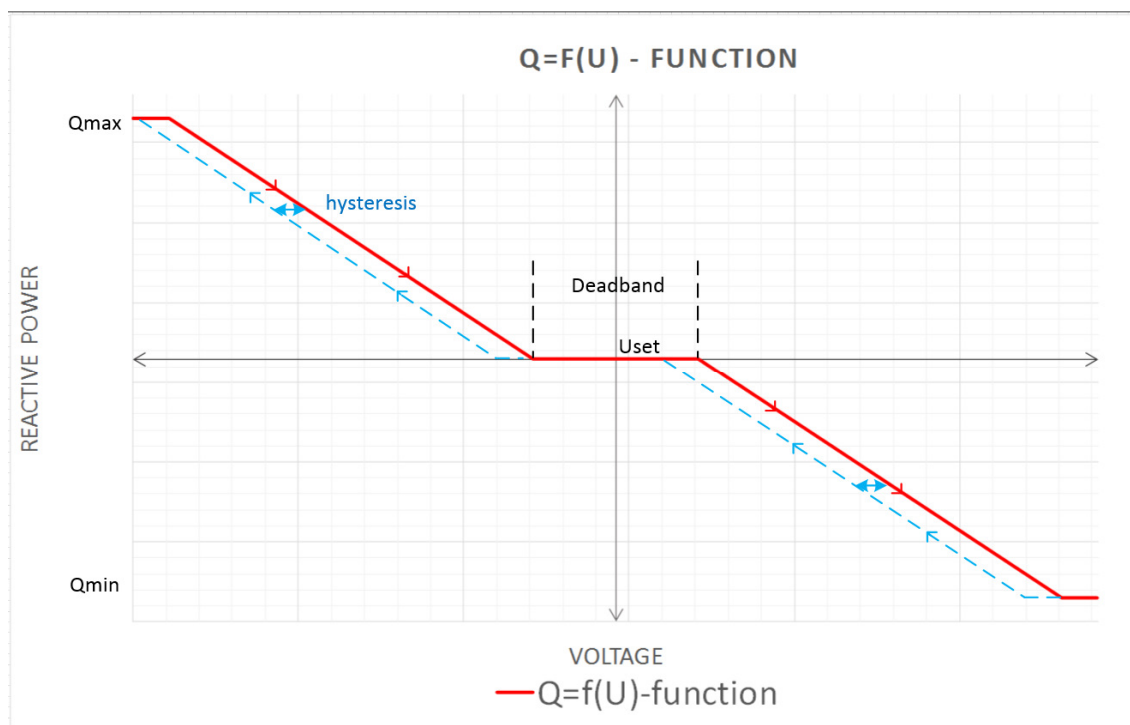


Figure 8: $Q=f(U)$ -function with hysteresis

	Sales Document Voltage Control Operation	K0821_045435_EN_R04 Revision 04 17.06.2019
---	---	---

2.5 Step function

When the measured voltage exceeds a configurable voltage range the $Q=f(U)$ -function can set the reactive power set point to the maximum ($Q_{\max}$) value, respectively to the minimum value ($Q_{\min}$). The maximum and minimum can either be set to the $Q_{\max}$ and $Q_{\min}$ of the $Q=f(U)$ -function or the maximum and minimum of the available reactive power in the windfarm. The step function can also be configured with a hysteresis (Figure 10).

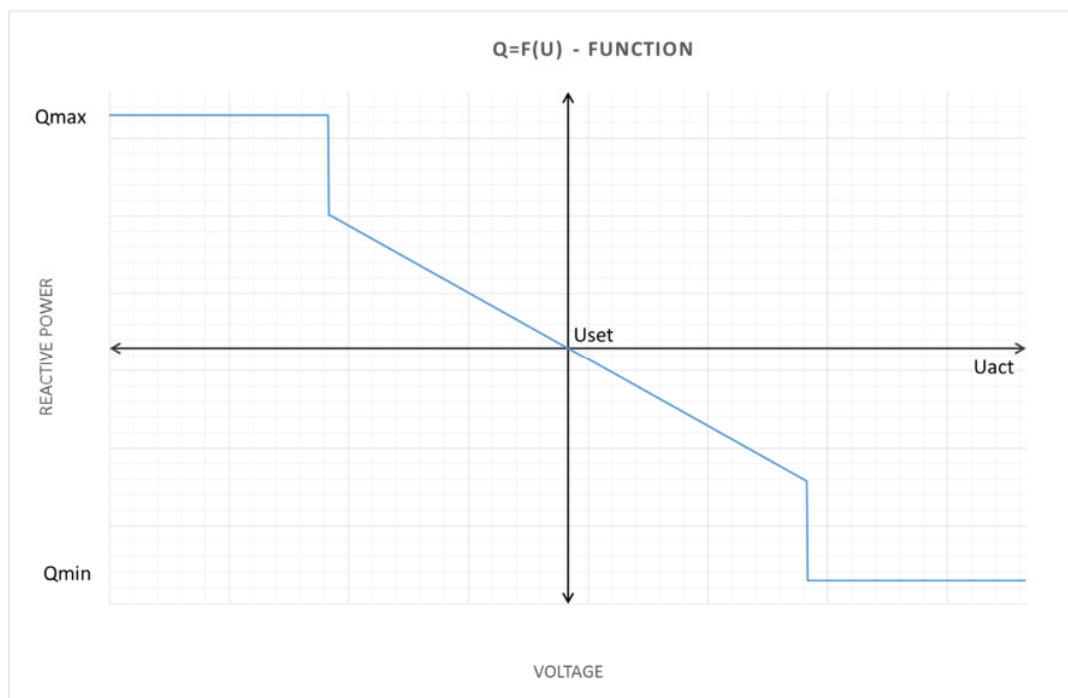


Figure 9: Step function

	Sales Document Voltage Control Operation	K0821_045435_EN_R04 Revision 04 17.06.2019
---	---	---

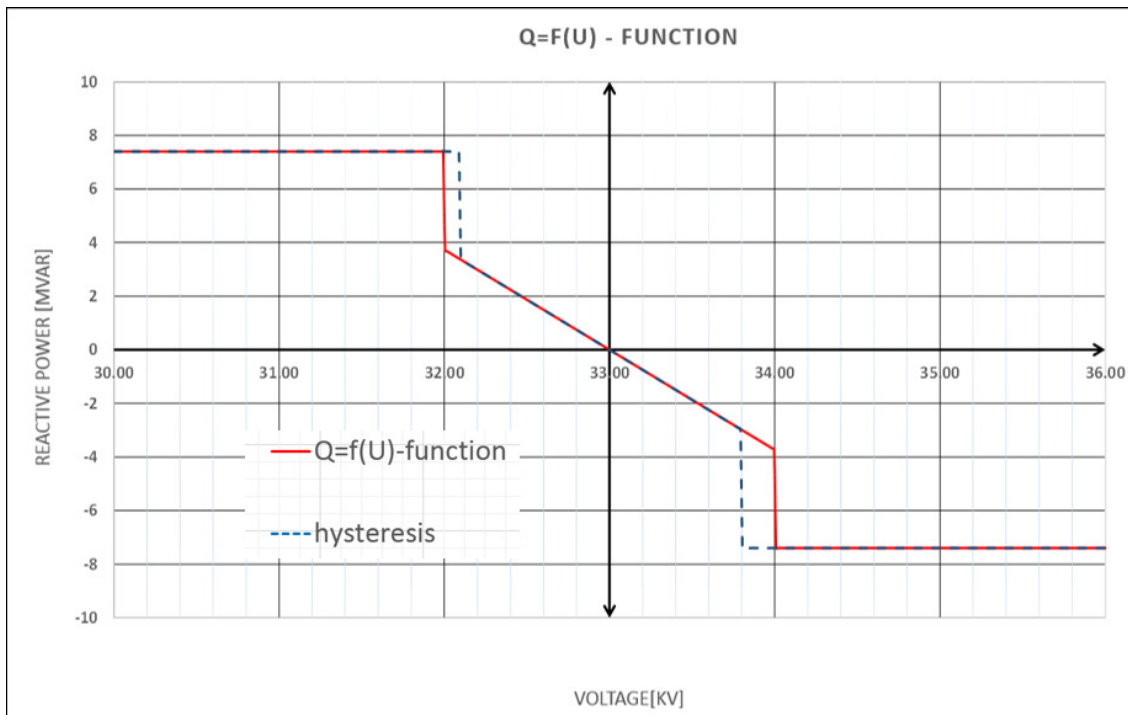


Figure 10: Step function with hysteresis

2.6 Reactive Power Capability

The general reactive power capability of the individual WTs has a direct influence on the reactive power capability of the wind farm and should be project specifically tested for this reason (refer to document K0801_004439_EN reactive power capability for K08, E0004514776 reactive power capability for Delta4000).

2.7 Set point Changes

The reactive power is changed when the voltage set point changes. The reactive power change occurs with a PT1-behaviour. The reactive power change effects a change in the voltage. The size and the course of the voltage change are dependent on the local grid conditions.

Figure 11 and Figure 12 show tests of a fast voltage control. Profinet-communication between the CWE and the WTs is essential for this. When these prerequisites have been fulfilled it will be possible for the step response to reach 90% of the set point within one second.

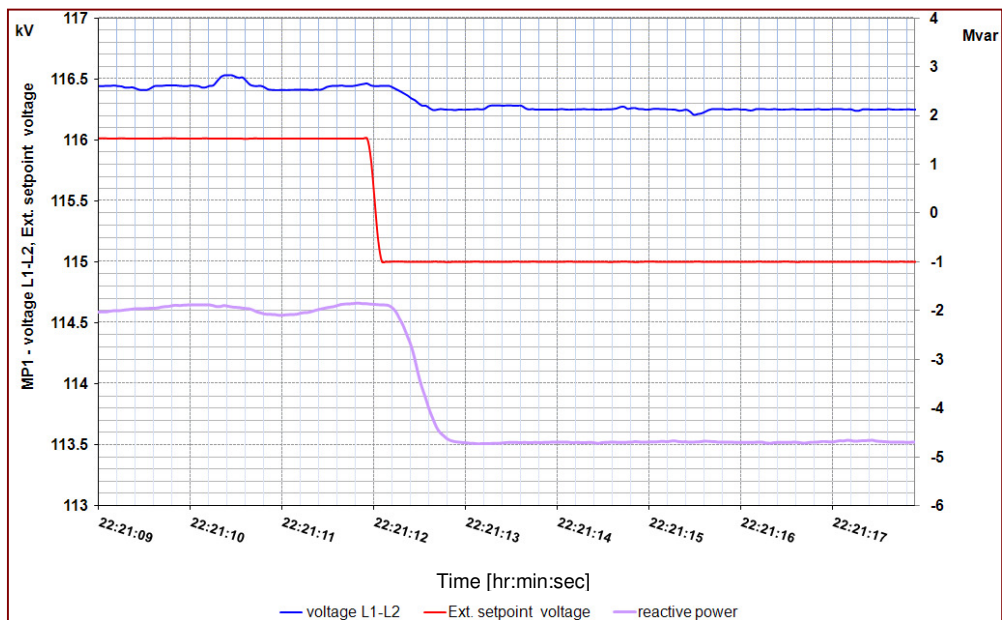


Figure 11: Field Measurement: Voltage set point jump from 116kV to 115kV

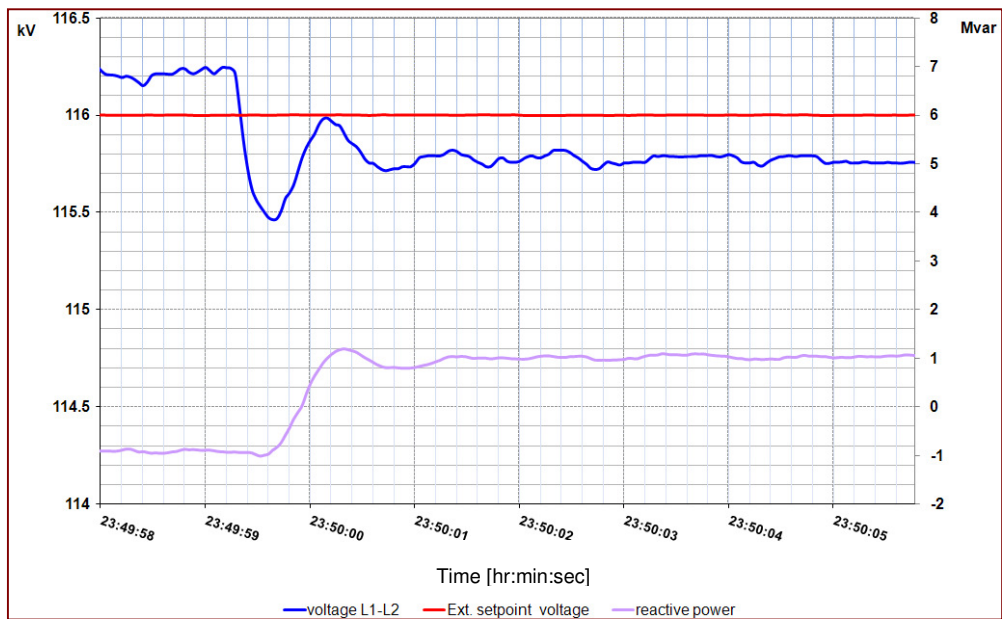


Figure 12: Field Measurement: Disturbance Variable Control

3 Measurement and control accuracy

	Sales Document Voltage Control Operation	K0821_045435_EN_R04 Revision 04 17.06.2019
---	---	---

3.1 Dynamic control accuracy

A general statement cannot be made on the control accuracy or the remaining control difference. These depend on the controller settings, the local grid characteristics and the current feed-in of active power.

3.2 Static measurement deviation

There is also a static measurement deviation comprising the following elements:

- Measurement transformer (Class 0.2 or 0.2s for the current transformer)
- Measurement transducer 0.2
- I/O-Assembly 0.06
- Quantization error

With all factors taken into consideration, this results in a measurement deviation of nearly 2 %.

3.3 Interfaces

The measurement point is the reference point for the active and reactive power and should correspond with the grid point of connection. These measurement values and the external set points are input to the CWE via one or more analogue inputs or a number of transducers, and are described in the signal list.

When required by the project, the set points can be defined via the visualisation or the 4..20 mA-interface. The 4..20 mA-interface set points are enabled via digital inputs and activated for the open- or closed loop control. The information on the measurement- and set point values can be output via analogue outputs for further processing and display

.

4 Protection Notice ISO 16016

The reproduction, distribution and utilization of this document as well as the communication of its contents to others without explicit authorization are prohibited. Offenders will be held liable for the payment of damages. All rights reserved in the event of the grant of a patent, utility model or design.